

Descriptive Information

1. Corporation # (if applicable)	2. Corporation Name/School Name/Organization Name University of Notre Dame		
3. Address (Street, City, State, Zip) Notre Dame, IN 46556			4. Telephone 574-631-5000
5. Contact Person's Name Dr. Thomas J. Loughran		6. Contact Person's Email Address Thomas.J.Loughran.8@nd.edu	
7. Contact Person's Address (Street, City, State, Zip) 225 Nieuwland Science Hall Notre Dame, IN 46556-5670			8. Contact Person's Telephone 574-631-3362
9. Superintendent/Principal/Director's Name Professor Mitchell R. Wayne		10. Superintendent/Principal/Director's Email Address Mitchell.R.Wayne.1@nd.edu	
11. Number of Students in Corporation or School There are over 100,000 K-12 Students in North Central Indiana.		13. Number of Teachers in Corporation or School There are over 4,000 K-12 Teachers in North Central Indiana.	



OFFICE OF RESEARCH

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November 10, 2011

Indiana Department of Education
Innovation Fund
Re: University Contributions

Dear Dr. Bennett:

We are pleased to participate in Indiana's first-ever Innovation Fund competition. We thank you and your staff for making this competition possible. As a sign of our enthusiasm for our proposal, The Notre Dame extended Research Community, we commit the following funds and resources to the 2-year project if it is awarded the requested state funding.

Department of Computer Science and Engineering	\$ 10,000
Department of Mechanical Engineering	\$ 10,000
Department of Electrical Engineering	\$ 10,000
College of Engineering	\$ 30,000
Department of Biological Sciences	\$ 10,000
Department of Chemistry and Biochemistry	\$ 10,000
Department of Physics	\$ 10,000
College of Science	\$ 30,000
Office of Research	\$ 120,000
The Graduate School	<u>\$ 120,000</u>
	\$ 360,000

In addition, the University will contribute \$380,642 in the use of facilities and administrative services, \$164,000 in graduate school tuition and \$100,000 in equipment. The total University contribution over the 2-years will be \$1,004,642. One-half of each of these contributions will be made each year of the project.

Sincerely,

A handwritten signature in blue ink, appearing to read "Liz Rulli".

Liz Rulli
Associate Vice President for Research

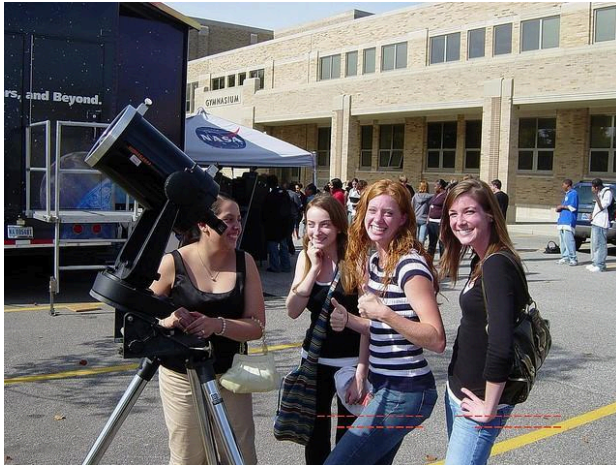
NDeRC

DOMAIN I: PROJECT DESCRIPTION

Part I: Brief Description

a. History

Thirteen years ago, physicists from the Notre Dame QuarkNet Center (NDQC) invited a dozen high school physics teachers to join with them in building the high energy physics community. Meeting weekly throughout the academic year and conducting full time research during the summer, we grew in our understanding of particle physics and began to take on the values of the research community. Teachers became research--focused, even as the researchers grew more aware of the opportunities and challenges in K-12 STEM education. Teachers naturally invited students from their own schools to participate with us in what began nominally as a program just for teachers. We continued to attend weekly meetings and conduct summer research together, year after year. Slowly it occurred to us that what we were engaged in was a different model of STEM education than any of us had considered before. And it seemed to be working.



Like other members of the scientific community around us, we took on commitments, forming collaborations with regional and national groups. At the national level, collaborations began with the emerging Research Experience for Teachers (RET) community, the Supercomputing Education Program, the National Optical Astronomy Observatory, and NASA. Locally, we began broader-impact initiatives with the molecular nanoelectronics group in Notre Dame's departments of Electrical Engineering and Chemistry. We planned

and executed long-term professional development activities with our collaborators, such as a three-week national workshop in 2002 sponsored jointly by the national QuarkNet project and the Arizona State University Physics Modeling program.

We believed early reports from RET programs that authentic research experiences for teachers produce some gain in student test scores. But our activities were not undertaken for the sake of student tests or even for the sake of students. We were designing and manufacturing optical decoding units for the CMS detector at the Large Hadron Collider, whose principles and operations we (teachers and students alike) absorbed whenever we could.

We had been invited and were now doing



science; so were the few students we were able to invite into our collaboration. Science education became all about issuing similar invitations to the rest of our students: some to join as researchers, but all of them as citizens to whom the scientific enterprise belongs. Several NDQC members started research classes at their high schools. Collaborations began regularly with informal education partners (such as Saint Joseph County Public Library's *Science Alive!*) and with other university outreach programs.

In an effort to substantially increase the surface area of our emerging and integrated STEM community, 15 NDQC teachers spearheaded a successful NSF GK-12 proposal, adding graduate students to our collaboration and forming similar collaborations in other disciplines across the university. Now in its fifth year of NSF--funded operation, the Notre Dame extended Research Community (NDeRC) incorporates graduate students into the integrated STEM community model, reaches out to connect with a broader range of K-12 teachers and students, and fosters collaborative relationships among institutional stakeholders.



perspective, STEM education does not seem to be about the kids, be they graduate students or 5th graders. It's about the research. Researchers need a steady flow of new participants to get the work done, and an informed and supportive public who values their work. Students—young researchers—need to be needed by the research community; they need to take ownership of the research; it belongs to them (and all of us) by right.

Among K-12 teachers, particularly, NDeRC's focus is on development of a sense of regional professional identity as STEM educators whose primary responsibility is to extend effective invitations to students to participate in STEM community. We extend these invitations first

Notre Dame QuarkNet Center

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June 14, 2006

Dear Committee Members:

We, the undersigned, are high school teachers and active members of the Notre Dame QuarkNet Center. Many of us have been members of the Center for more than five years. As members of the Center we have engaged in particle physics research alongside physicists, engineers, technicians and over 100 high school students from our schools. The bulk of this work is done during the summer, but we continue to meet weekly during the academic year so as to maintain the continuity of our culture and community.

We are very excited about The Notre Dame Extended Research Community (NDeRC) proposal. We are especially excited about the prospect of interacting with graduate students in a substantial and continuing manner. We believe that graduate students would enrich our culture and we theirs. This proposal gives us a venue for exchanging technical and pedagogical expertise and support: during the summer, during the academic year, in our classrooms and in our laboratories.

Patrick Mooney
Patrick Mooney, QuarkNet 7-years
LeRoy Castle
LeRoy Castle, QuarkNet 7-years
Jeff Chorny
Jeff Chorny, QuarkNet 6-years
Tom Guthrie
Tom Guthrie, QuarkNet 6-years
Lynda Rose
Lynda Rose, QuarkNet 3-years
Aaron McNeely
Aaron McNeely, QuarkNet 2-years
Helene Deurty
Helene Deurty, QuarkNet 6-years

Tom Loughran
Tom Loughran, QuarkNet 7-years
Dale Wiand
Dale Wiand, QuarkNet 7-years
Ken Ander
Ken Ander, QuarkNet 6-years
Calvin Swartzendruber
Calvin Swartzendruber, QuarkNet 3 years
Stacy McCormack
Stacy McCormack, 2-years
Mark Kirzeder
Mark Kirzeder, QuarkNet 1-year
Richard D. Eberly
Richard Eberly, QuarkNet 2-years

6/16/2006

b. Mission

NDeRC activities became increasingly focused on building an integrated STEM community. STEM education, on this model, just is the issuing of effective invitations into STEM community. Doing authentic research is not seen as a strategy for getting STEM content to stick; rather, content is mastered so that research can advance. From this

to teachers, and with them, to students, by laying out a broad spectrum of engagement opportunities for teachers across the full K-12 spectrum.

The annual Collaborating for Research and Education Forum (now approaching our fifth year) has been the central hub for our invitational activity. At the Forum, we present a broad spectrum of opportunities, including as many as we can from non-NDeRC sponsors. Invitations to participate are issued, and barriers to sign-up are lowered. K-12 teachers make up the largest number of participants in these events; we carve out space in the schedule to listen to their concerns. The Forum's central task is always to foster a vibrant STEM community with a rich STEM culture.

This strategy is pursued in a number of interacting but autonomous collaborations: BioEYES, NANO, ENVIRO, GENO, ASTRO, Particle Physics and Computational Science. Collaborations are spearheaded by teams each consisting of one or two graduate students, one or two lead teachers, and one or two high school students, working under the guidance of a faculty advisor. Each team's summer-long research activities are leveraged to create introductory experiences in that area of research for new K-12 teachers through week-long Institutes offered in successive summers. Each Institute is an invitation into research: preferably, to one or more summers of research. Core concepts are explored through hands-on activities that will be brought to each participating teacher's classroom by a graduate student and/or an NDeRC staff educator. Participating teachers engage in those classroom activities themselves in the Institute, visit laboratories where research in each area is conducted, and learn about the lead team's research activities. Momentum from the Forum, the Institutes, and the week of classroom transfer (STEMweeks) is maintained through biweekly meetings throughout the school year, together with blogging by collaboration members. The Forum, Institutes, STEMweek, biweekly meetings and online collaboration together make up the ordinary pattern of life of NDeRC.

The intent behind all these activities is to invite teachers and students into integrated STEM community, but in ways respond to the specific needs of teachers who have responsibilities to students as defined in K-12 schools. So, STEMweek activities are aligned to state standards, for example; language arts instruction is incorporated into Institute activities, some of which are tailored for specific grade levels. In these ways, the collaboration is acting as a good neighbor to classroom teachers who have responsibilities under K-12 educational guidelines. But the STEMweek activities do not exist primarily to advance the business of school. Their purpose is to invite teachers—and together with them, to invite students—into STEM community.

c. Prior Achievements



NDQC continues to meet weekly, with no direct financial recompense for participating teachers, throughout the academic year. After 13 years, longevity is some evidence of community and thus proof-of-concept. Generally, members of NDeRC who teach physics and some who teach chemistry are members of NDQC and participate in those weekly meetings.

Detector research, design and manufacture continue to advance at a faster pace because of teacher and student participation: 500 optical decoding units now process signals from collisions at 25 million events per second at the world's highest energy collider...all assembled by high school students who understood their structure and function. One pair of student researchers investigating material properties for this construction effort became regional finalists in the Siemens Science Fair competition.



New Prairie High School biology teacher John Gensic is one of five Jacobs Educators for the Indiana University School of Education and has received stipends to enhance science activities. (South Bend Tribune Photo/GREG .SWIERCZ / September 2, 2011)

NDQC members continue to advance as educational entrepreneurs. On account of her educational entrepreneurial efforts, PI Mitch Wayne successfully nominated Stacy McCormack as the 2011 Indiana Teacher of the Year. John Gensic, an ACE graduate now teaching at New Prairie High School, a veteran of three or more NDeRC institutes and a lead teacher with multiple years of research experience, was successfully nominated by NDeRC as a 2011 Jacobs Educator at IU School of Education, Bloomington. Sharon Brandt, a teacher at South Bend Community School Corporation's LaSalle Intermediate

Academy, successfully proposed a 1-hour workshop on bringing Nanotechnology to K-12 classrooms for the upcoming National Science Teacher's Association conference to be held in Indianapolis next Spring.

Four annual Collaborating for Education and Research Forums have been held thus far. (345 K-12 teachers have attended at least one; 143 of them at least two, 63 at least three.) Forum participants are in near unanimity (100% unanimity last year) that the Forums succeed in advancing their goal of inviting participants into collaborative relationships that foster vibrant STEM community and rich STEM culture.



Over the past four years of NDeRC's operation, three colleges, six departments, 14 T&R faculty and 19 graduate students have participated with NDeRC in extending invitations to some 325 teachers and more than 13,000 students in Michiana. (These numbers reflect only NDeRC's own programs, leaving out a great many activities of other collaborators to which NDeRC added value.) In the past year alone, NDeRC STEMweeks have been in 260 different classrooms with 100 Michiana teachers and 6600 students in 70 different schools, roughly two thirds of which were Indiana public schools, more than 95% being Indiana schools.ⁱ

NDERC K-12 Classroom Activities 2010 - 2011

Subdiscipline	Collaboration	# Teachers	# Students
developmental biology and genetics	BioEYES	62	4020
nanoscience and technology	NANO	14	1006

environmental engineering	ENVIRO	11	977
molecular biology	GENO	4	198
astrophysics	ASTRO	4	180
particle physics	CMS Dataclass	2	130
computational science	Alice/MATLAB	2	50

Part II. Project Plan:

a. Project Need

The academic achievement and career preparation of all Indiana students will be the best in the United States and on par with the most competitive countries in the world. ~*Indiana Department of Education Vision*

K-12 education in North Central Indiana has come under a great deal of criticism, some of it from the State Department of Education. Three South Bend Community School Corporation high schools nearly failed to make the adequate yearly progress on standardized tests necessary to escape state intervention. These prominent symptoms of need mask deeper problems. Noted Harvard educator Tony Wagner summarizes the theme he repeatedly encountered in interviews with top practitioners in the new knowledge economy: even in good schools...

“...students are simply not learning the skills that matter most for the twenty-first century. Our system of public education—our curricula, teaching methods, and the tests students are required to take—were created in a different century for the needs of another era. They are hopelessly outdated.” (*The Global Achievement Gap*, 2008.)

The central need, here and elsewhere in Indiana, is for teachers who are equipped to lead today's students: teachers who exercise 21st Century entrepreneurial skills that their students will need to acquire if they are to live well as citizens in a global knowledge-based economy.ⁱⁱ The State of Indiana DOE has implicitly recognized this need in their advocacy for the New Tech High (NTH) network. The hope is that the introduction of NTH schools throughout Indiana districts will catalyze the spread of 21st Century project-based learning (PBL).ⁱⁱⁱ

The challenge facing Indiana and indeed all U.S. K--12 education is to identify and implement a different way of educating children, like New Tech High, while at the same time not abandoning the children under the care of the present system. There is some tension between the call to 21st Century learning and the need for accountability in doing better what schools already do: the standard of achievement is not the same, and arguments for scarce resources will begin from different premises. That tension seems mitigated to some extent by the track record of successful performance on standardized tests for PBL students, but this is peace only from the perspective of schools.

A more profound tension exists not regarding student achievement but regarding teacher professional development. “How can I teach the seven survival skills to students when I haven’t learned them?” With this common question teachers ask of Tony Wagner, he closes his 2010 afterword to his provocative masterwork, *The Global Achievement Gap*. To what extent, Wagner asks, is a culture of innovation a precondition for the new kind of learning for adults and students alike?

Wagner’s choice of culture is an appropriate focus. As recent work on collective impact suggests, no piecemeal approach, no single programmatic fix, will suffice to produce the kinds of 21st century social entrepreneurs—the adults—who will be able to guide students into the new adulthood.^{iv} Nothing less than a cultural transformation will do. Culture makes what is good, look good, not from a single fleeting perspective but from nearly every perspective. Culture leads us in the way we should go. We need teachers who see from nearly every angle that the world is theirs to transform, empowered as they are by a host of collaborative opportunities and supporting technologies. A culture of innovation, supporting social and commercial entrepreneurship, must be an enduring and wide-scale collective impact. It can only be the product of the efforts of many persons acting in the context of a broad spectrum of engagement opportunities with a wide range of community partners. Though it will not make the morning papers, this is Indiana’s central educational need.

b. Project Objectives

NDeRC’s goal of fostering integrated STEM community and rich STEM culture is the target our community needs to hit in order to hasten the coming of 21st Century educational reform. The central goal is the same for everyone: active participation in a community whose defining activity is collaborative research in the STEM disciplines. For the purposes of this proposal, we draw a distinction between leaders and other collaborators for both K-12 students and teachers. The pattern is fairly simple: first we meet one another, then work together (acquiring skills and knowledge as needed), and finally produce results together.

Participant type	Short-range goals	Mid-range goals	Long term goals
K-12 collaborating teachers	Collaborative relationships ¹	Collaboration skills ²	Successful collaborative activity, values ³
K-12 collaborating students	Collaborative relationships ⁴	Collaboration skills, content knowledge, (language arts success) ⁵	Successful collaborative trajectories
K-12 Lead Teachers	Research and collaboration opportunities ⁶	Research and collaboration experience ⁷	Collaborative research productivity, values ⁸
K-12 Lead Students	Research and collaboration opportunities ⁶	Research and collaboration experience ⁷	Collaborative research productivity, values ⁸
	Academic community	Academic community	Successful academic

ND STEM Faculty	engagement opportunities	engagement relationships	community engagement experience
ND STEM Graduate students	Collaborative leadership, research communication opportunities	Collaborative leadership, research communication experience	Collaborative leadership, research communication success, portfolios
ND Undergrads with STEM competencies	Community engagement, STEM research opportunities	Community engagement, STEM research experience	Community engagement leadership, STEM research success

Orange: assessment key for this project (see Domain II, Part I d for details); Red: measured under this proposal; Black: background goals not assessed as part of this project; Blue: column and row headers

c.

Project Description

Our pattern of activity—Forum, Summer research, Institutes, STEMweeks, and the relationship among these—has been described in general terms in the Mission section above. Here we provide collaboration details for just the STEMweek activities: summer research topics for each

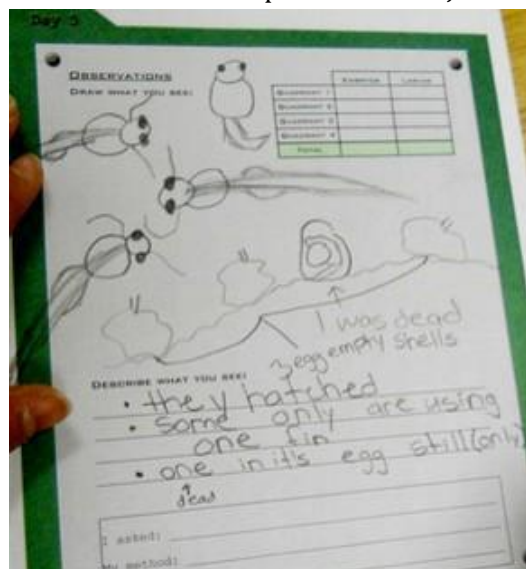
collaboration can be gleaned from collaboration blogs and from faculty advisors' home pages. We also include a section on new developments under this proposal.



Collaborations



The **BioEYES^v** collaboration offers a week--long activity using zebrafish as a model to give students experience with the fundamentals of embryonic development and genetics in their own classrooms. These week--long classroom activities at primary, middle and high school levels were developed at Thomas Jefferson University and were introduced to us by Professor David Hyde^{vi}, the Director of Notre Dame's Center for Zebrafish Research. Zebrafish are a popular genetic model for research scientists because they develop quickly, are transparent during development, have a genome that is easy to manipulate, and are cost effective. At every level of the activity, students breed zebrafish, observe their embryos developing from a few cells to swimming fry in three days, and notice genetic differences in the offspring the students themselves have produced. This is an equipment--intensive activity, spearheaded by faculty and graduate students but normally guided by our BioEYES educator, Anita Beebe, supported by a van and a dedicated quarantine rack in the Friemann Life Sciences Center^{vii}. (See the links^{viii} to blogs and Faculty Advisor David Hyde's home page.)





NANOweek introduction of Scanning Tunneling Microscopy at Swanson Primary Center: students taking data (left); building a model (center); deploying the model using STM instrumentation with ND grad students (right.)

NANO is our nanoscience/nanotechnology collaboration. Classroom activities are focused on getting a sense of scale and on the technological wonder of seeing the very small. The project introduces the electromagnetic spectrum, has a heavy emphasis on units and measurement, and promotes understanding and appreciation for measurement instrumentation. The hands-on activities particularly at the younger grades have a cultural value, rebranding the idea of nanotechnology as a positive and fun activity. (The children pictured in the series of images above are first graders.) Research in the collaboration is centered around molecular nanoelectronics; versions of the same instrumentation that campus researchers use to explore the world at the nanoscale are brought into classrooms, supported by NDeRC's second van. The collaboration has functional traveling Scanning Tunneling (two of these), Atomic Force, and Scanning Electron microscopes. Faculty advisors^{ix} Gary Bernstein (EE), Alex Kandel and Marya Lieberman (Chem/Biochem) support year-long high school student research.



Our **ENVIRO** collaboration has developed two distinct flavors. The first of these focuses on well water testing and mapping; 143 wells were tested and mapped by Indiana middle and high school students last year. Stephen Silliman^x (CE/GS) serves as faculty advisor for the wellhead mapping effort. The second effort is a sociological study of student attitudes toward their environment in a school gardening project at South Bend Community School Corporation's Marquette Montessori magnet.

Collaboration blogs and wikis are linked here.^{xi}

The Michiana Astronomical Society's preparations for the 2012 transit of venus are the focus of this year's **ASTRO** activities. A mashup of free online tools where anyone can create astroimages by mining the world's astroimaging databases is a hit both in the summer Institute and with middle school students. It is worth mentioning that members of the summer research group have become Minor-Planetary-Center-certified asteroid hunters, and were granted preliminary discovery notices on three new asteroids during Summer 2010. Bremen High School teacher Aaron McNeely and SBCSC Adams High School's Carolyn Fletcher lead these student research teams. The ASTRO faculty advisor is Chris Howk. Follow links to that activity as well as ASTRO blogs and wikis here.^{xii}



Our **GENO** collaboration is developing ways to export to classrooms the molecular biology techniques we first introduced to teachers during the BioEYES institute. Working with South Bend born start-up Genotyp, Inc. (formerly Peyer Labs), the collaboration works to fine-tune Genotyp's Cloning a Fluorescent Gene activity. Faculty advisor: Holly Goodson; links are here.^{xiii}

Our **Particle Physics** and **MATLAB** (computational science) collaborations are

smaller and depart from the Institute-STEMweek model, though both have graduate student, high school teacher and student research components. Mike Hildreth (Phys) is Faculty Advisor for particle physics; Craig Lent (EE) for MATLAB. Links are here.^{xiv}

New Initiatives

Undergraduate participation in an otherwise well-integrated research community has been fairly minimal until now. (Undergraduate support was prohibited under NSF GK-12 program guidelines.) But two winds blow in favor of undergraduate participation at Notre Dame: a strong service-learning tradition across the University; and an emerging commitment to undergraduate research. Our experience with NDeRC is that it is possible to create additional lines of research with high school students and teachers during the summer, then continue them with undergraduates during the academic year. (This approach already works for high school academic year research students who work either year-round or inherit projects begun in the summer by other students.) We intend to look for ways to increase the surface area of University research. We will work synergistically with graduate students, undergrads, high school teachers and high school students, by adding undergraduate students to our existing graduate/high school research teams. In addition to increasing the available active possibilities for research, undergraduate students could buffer the demands on graduate student

research time during the summers, and could be part of the teams visiting schools during STEMweeks. Co-PI Loughran, who conducted and published undergraduate research at Notre Dame during a summer and a calendar year and now leads a research class at a local high school, will spearhead the effort to involve undergraduates in NDeRC collaboration research. Moreover, in his capacity as Education

Program Leader for I2U2, Loughran has designed online research environments, a dimension of the research space that might further open up avenues of interaction between undergraduates and K-12 teachers and students in authentic research experiences. Loughran will work with Co-PI Brockman, who has coordinating responsibilities for project selection for College of Engineering graduate fellows.



South Bend New Tech High School Principal John Kennedy and NDeRC have agreed that all NHT teachers have preferred access to lead--teacher and other professional development activities, consistent with previous NDeRC commitments. Co-PI Brockman, already the NTH liaison with the University of Notre Dame, will continue to advise NTH and coordinate interactions with them.

NDeRC proposes to add **scaffolded blogging on standards-based topics** to STEMweek activities, beginning in year one with BioEYES and moving into other STEMweeks in year two. Particularly in the BioEYES collaboration, NDeRC promotes writing on STEM topics at all K-12 grade levels on paper; this year we propose to carry a small fleet of netbooks for student (internet or intranet) blogging. An example of a scaffolded blogging opportunity is provided below. We will ask students to write in ways that show their grasp of a given standard to their funding partner—the State of Indiana—while paired in teams as “content accountability peers”. Each student writes a blog post, which their partnering student reads and comments on, giving a net “yea” or “not yet” for each standard assigned; a “thumbs up” icon will be activated for each “yea”.

Indiana State Standard		Assignment	Key terms
BioEYES: Grade 7	7.4.3 Explain how, in sexual reproduction, a single specialized cell from a female merges with a specialized cell from a male and this fertilized egg carries genetic information from each parent and multiplies to form the complete organism	Write a creative but factually accurate representation of your week studying the life of zebrafish.	* fertilized egg * genetic information
BioEYES Grade 8	8.2.8 Use tables, charts, and graphs in making arguments and claims in, for example, oral and written presentations about lab or fieldwork. 8.4.1 Differentiate between inherited traits, such as hair color or flower color, and acquired skills such as manners. 8.5.6 Explain that a single example can never prove that something is always true, but it could prove that something is not always true.	Explain whether or not your initial hypothesis was proven or disproven. Support your argument with Punnett squares and/or data gathered and displayed properly in the form of bar graphs.	*hypothesis *genetic information *inheritance of traits

Students will be shown how to incorporate images of their fish in ways that school internet use policies permit. Blogs thus become artifacts for assessment; numbers of posts paired with thumbs up can be counted, and student-rater reliability can be determined by sampling and applied to large numbers of posts. In this way, frequency data can be gathered, together with crowd-sourced (student-sourced) semantic analysis, a possibility we would like to explore as we move toward greater amounts and kinds of online interaction.^{xv} An additional benefit of scaffolded blogging is enabling the BioEYES invitation to reach a larger number of persons, online: where policies permit, blogs after review will migrate from intranet to the web.

Collaboration technologies (C21) training will be provided by Loughran, a Google Certified Innovator, and by Pam Burrish, ACE Coordinator for Technology Outreach, also a GCT and Apple Distinguished Educator. Chris Clark, Assistant Director and Learning Technology Lab coordinator at Notre Dame's Kaneb Center for Teaching and Learning, will advise. In a week's time, teachers will develop a coordinated online presence (with integrated LinkedIn, Skype, Twitter, Flickr, Facebook, and Google Apps accounts) and begin an online portfolio of STEM collaborative activities. The purpose of this addition is to enable collaborations to connect with one another more efficiently and be reflected online so that others might benefit or even join.

d. Project leaders

We present here in tabular form the project leaders. Curriculum vitae are attached.

Name	University Position/Role	Project Role	Responsibilities
Mitchell R. Wayne	Professor/Department of Physics, Chair/Formal Associate Dean of Science	Principal Investigator	Strategic Planning, Budgeting, Relations with State and University; Oversight
Jay B. Brockman	Associate Professor/Department of Computer Science and Engineering & Department of Electrical Engineering/Associate Dean of Engineering for Educational Programs/	Co--Principal Investigator	Strategic Planning, Coordinate Research Project Selection for Engineering, Coordinate with South Bend New Tech High
Karen M. Morris	Professional Specialist/ Department of Chemistry and Biochemistry/Center for Sustainable Energy at Notre Dame – Education and Outreach Program Manager/Inst for Educational Initiatives – Fellow and Science Consultant in the Office of Educational Outreach	Co--Principal Investigator	Strategic Planning & Coordinate Relations with Schools and School Districts
Thomas J. Loughran	Professional Specialist/Department of Physics/NSF GK--12 Project Managing co--PI/I2U2 Education Program Leader/Google Certified Teacher/High School Research Class Teacher	Co--Principal Investigator	Strategic Planning, New Initiatives, Collaboration Technologies & Assessment, Coordinate Undergraduate activities
Patrick J. Mooney	Professional Specialist/Department of Physics/NSF GK--12 Project Coordinator	Co--Principal Investigator	Strategic Planning, Budgeting & Execution
Rachel Fulcher Dawson		External Evaluator	To assess and evaluate the projects progress towards its goals, to report findings to the project management and offer management informed input on possible modifications in approach

Domain Two: Impact on Student Achievement

Part I: impact on student achievement.

a. Number of students impacted

As their teachers become ready to lead them as 21st Century social and commercial entrepreneurs, the impact on student quality of life will be profound. Our project aims to build community first with teachers. Social network analysis enables frequency counts, but only limited semantic analysis of our impact on teachers, let alone students. We propose to do both: count the frequency of contact, and observe and record meaning. Our scaffolded blogging activity will begin to give us some way to gather semantic data on scale, but it will just begin. Counts of collaborative activities forecast for the next two years are below.

Collaboration	Year 1	Year 2	Total years 1 & 2
BioEYES	5500	6500	12,000
NANO	1000	1500	2500
ASTRO	200	300	500
ENVIRO	500	1000	1500
GENO	200	300	500
TOTALS	7400	9300	17,000

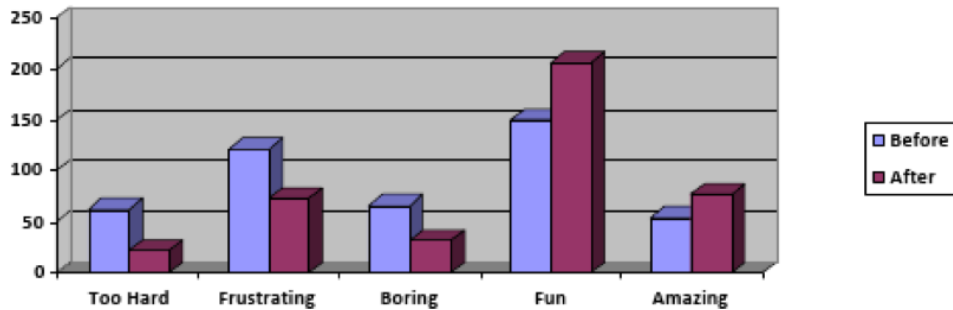
We expect after year one to have at least 1000 blog posts for analysis.

b. Research-based approach?

NDeRC is an educational entrepreneurial initiative. In spite of the length of time we have been working together, we seem to be implementing a distinctive model. From the literature we learn the value of making a collective impact; we are not yet organized so broadly as Strive in Cincinnati. We have been building STEM community and establishing rich STEM culture, goals which arise more from our experience than from research.

c. Evidence of success?

Our approach begins with teachers; from our forum counts and surveys it is clear that we are increasing the number of opportunities for teachers to collaborate. After stipending teachers for 3 years running to attend the forum, we offered no stipend in year four and still drew over 100 K-12 teachers to the event. This is our best evidence of success. Our STEMweek activities for students are designed as ways for teachers to share with their students their emerging connections in STEM outside the classroom. We design them to be fun. We have sampled the efficacy of our BioEYES activity: when asked both before and after the week-long classroom BioEYES activity to circle "five words (of twenty options) that best describe (their) thoughts and feelings about science", a limited sample of the responses of participating girls (n=332 girls in middle and high school) indicates positive movement in student attitudes toward science:



d. What data will be collected?

The following data sets are keyed to the participant outcomes listed in the project objectives section:

¹ #Forum attendance; #Institute attendance; #blog subscriptions

² Institute posts; C21 profiles and portfolios

³ C21 portfolio entries; repeated forum attendance

⁴ number of students reached; # pairs of blogs/comments made

⁵ #pairs of blogs/comments verified

⁶ research blogs & wikis (activity reflected online): # of posts

⁷ blogs and wikis analyzed

e. Evidence for implementation of new elements?

There is evidence that blogging and scaffolding treatment can help improve student narrative writing ability (in a grade five study.) ^{xvi}

f. Benefits of expanding?

Adding undergraduates to stabilize additional research threads promises to increase the surface area of research opportunities at the University. Adding the scaffolded blogging element promises to increase the visibility of BioEYES, a good outcome for any activity designed to grow a community. C21 training for teachers will have a similar effect, and will make it easier to disseminate the integrated STEM community model.

Part II: continuous improvement of the program.

a. Systems to ensure academic progress?

NDeRC is not directly involved with measuring academic progress.

b. Mechanisms for creating change?

NDeRC is adding two very experienced PIs (Brockman and Morris) who have deep personal investments in K-12 education. We invited them as PIs in part because of the constant flow of formative assessment we expect to receive from them.

c. Sustaining the program?

Co--PI Brockman is co--chair of the newly established Provost's standing committee on academic community engagement. Their charge is to recommend a strategy for institutionalizing academic community engagement at Notre Dame. The large and broad--based commitment of matching funds for this proposal is some indication of a genuine

desire at ND to find a successful engagement strategy; NDeRC's integrated STEM community approach will be among those considered.

Also, dissemination is one sense of sustaining a program. The Saint Joseph County Public Library, University Libraries, and the Center for Social Concerns are in discussion with us concerning developing a collaboration space and a trained staff to catalyze 21st Century community--based projects. Libraries represent a network of potential collaboration nodes throughout the state of Indiana. We look forward to developing this new initiative.

Domain Three: Budget and Timeline

a. The integration of the local STEM community will advance as its members desire and strive for a common future: common life and common work. This suggests that its members plan for and participate in communal activities that are both regenerative (life) and rewarding (work). All the activities in the project timeline are designed to be communal and collaborative – to regenerate and reward. The budget which supports these activities is an essentially equal combination of State (\$999,434) funds and University (\$1,004,642) funds and contributions.

The University funds will support the following:

1. \$269,840 for Graduate Fellow supplemental stipends, benefits, training and travel
2. \$60,624 for Undergraduate Fellow stipends, benefits, training and travel
3. \$21,536 for a portion of the Integrating Coordinators salary and benefits
4. \$8,000 for Integrating Coordinator and Project Coordinator travel

These expenditures are essential for the Fellows and Coordinators to fulfill their project responsibilities as outlined in the timeline. In addition to the above \$360,000 in funds, the University will contribute \$380,642 in the use of facilities and administrative services, \$164,000 in graduate school tuition and \$100,000 in equipment for a total of \$1,004,642 in University funds and contributions over the 2--year period. One--half of each of these contributions will be made each year of the project.

i <http://tinyurl.com/nderc2011master> is a shared copy of the master list of the 2010--11 STEMweek schedule. ii The case for this need, not just in Michiana but world--wide, has been well assembled in "High Performing Districts in the Application of 21st Century Learning Technologies: Review of the Research", Friesen and Lock, for the College of Alberta School Superintendents: http://o.b5z.net/i/u/10063916/h/Pre--Conference/cass_lit--review_final.pdf. But Tony Wagner's *The Global Achievement Gap*, Basic Books, 2008, Afterward 2010, is a readable and popular case. iii Project--based learning is well described in "A Review of Research on Project--Based Learning", John Thomas, AutoDesk Foundation March 2000, available at http://www.ri.net/middletown/mef/linksresources/documents/researchreviewPBL_070226.pdf. For a timeline of CELL's successful efforts to introduce NTH in Indiana, see <http://cell.uindy.edu/NTHS/newtechhighinindiana.php>.

iv "Collective Impact: STRIVE Cincinnati", B. Kania and M. Kramer, Stanford Social Innovation Review, 2011, http://www.ssireview.org/articles/entry/collective_impact/.

v <http://www.bioeyes.org>

vi <http://biology.nd.edu/people/faculty/hyde/> Professor Hyde has agreed to provide a graduate student from his or a colleague's lab and to serve in the Mentor's role for the BioEYES collaboration.

vii <http://nd.edu/~ndflsc/flscintroduction.html> Friemann hosts NDeRC BioEYES zebrafish (~1000) without charge.

viii collaboration blog: <http://erc.nd.edu/blogs/bioeyes/>; BioEYES educator Anita Beebe's blog:

<http://erc.nd.edu/blogs/abeebe/>, Faculty Advisor David Hyde's home page:

<http://biology.nd.edu/people/faculty/hyde/>. BioEYES collaboration wiki: <http://bioeyes.michianastem.org> ix

NANO Faculty Advisors Marya Lieberman: <http://www.nd.edu/~mlieberm/>; Alex Kandel:

<http://chemistry.nd.edu/faculty/detail/skandel/>; GARY BERNSTEIN:

<http://xml.ee.nd.edu/faculty/faculty.xsp?id=676265726e737465>; COLLABORATION BLOG:

[HTTP://ERC.ND.EDU/BLOGS/NANO](http://erc.nd.edu/blogs/nano); COLLABORATION WIKI: [HTTP://NANO.MICHIANASTEM.ORG](http://nano.michianastem.org); A SUMMARY OF THE HIGH SCHOOL RESEARCH IN ONE FACULTY ADVISOR LAB: [HTTP://ERC.ND.EDU/BLOGS/NANO/2011/05/26/A--SUMMARY--OF--MY--HIGH--SCHOOL--RESEARCH/](http://erc.nd.edu/blogs/nano/2011/05/26/a--summary--of--my--high--school--research/)

x ENVIRO FACULTY ADVISOR STEVE SILLIMAN: [HTTP://WWW.ND.EDU/~SILLIMAN/](http://www.nd.edu/~silliman/)

xi ENVIRO wiki: <http://enviro.michianastem.org>; blogs: <http://erc.nd.edu/blogs/enviro>

xii ASTRO Faculty Advisor Chris Howk: <http://www.nd.edu/~jhowk/>; ASTRO Collaboration wiki:

<http://astro.michianastem.org>; astroimaging activity: <http://astro.michianastem.org/astroimaging>; transit of venus collaboration, links: <http://erc.nd.edu/blogs/blog/2011/01/26/planning--for--transit--of--venus--2012/>; asteroid discovery blog post: <http://erc.nd.edu/blogs/mamcneely/2011/09/11/nderc--asteroid--research--summer--2011/>

xiii GENO Faculty Advisor Holly Goodson: <http://chemistry.nd.edu/faculty/detail/hgoodson/>; collaboration wiki: <http://geno.michianastem.org>; collaboration blog: <http://erc.nd.edu/blogs/geno>; Genotype website: <http://speakscience.org>

xiv Particle Physics Faculty Advisor Mike Hildreth: <http://physics.nd.edu/people/faculty/michael--hildreth/>;

MATLAB Faculty Advisor Craig Lent: <http://www.nd.edu/~lent/>

xv HERE WE PROPOSE TO TAKE A SMALL STEP TOWARD A BLENDED MODEL OF SEMANTIC ANALYSIS OF ONLINE COMMUNITY BY CROWDSOURCING. SEE "ANALYZING ONLINE TEACHER NETWORKS: CYBER--NETWORKS REQUIRE CYBER--RESEARCH TOOLS", SCHLAGER, ET AL., (2009) JOURNAL OF TECHNOLOGY EDUCATION, 60(1), 86--100.

xvi HEW, ET AL., "WEB 2.0 TECHNOLOGIES IN K--12 SCHOOL SETTINGS: EVIDENCE--BASED PRACTICE?" IN *ENHANCING LEARNING THROUGH TECHNOLOGY. EDUCATION UNPLUGGED: MOBILE TECHNOLOGIES AND WEB 2.0*, [COMMUNICATIONS IN COMPUTER AND INFORMATION SCIENCE](#), 2011, VOLUME 177, 319--328.

[HTTP://WWW.SPRINGERLINK.COM/CONTENT/N7261108807988h7/FULLTEXT.PDF](http://www.springerlink.com/content/N7261108807988h7/fulltext.pdf)

